WORKED SOLUTION

STEP I 2014, Question 2

Pure • Integration of Logarithms

STEP I 2014, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s1-q2

The key idea

Everything rests on factorising $x^2-4=(x-2)(x+2)$, so that a logarithm of a quadratic splits into $\ln(2-x)+\ln(2+x)$, two pieces that part (i) (and its mirror image) integrates at once. The modulus signs then only ask us to keep track of where $\ln|x^2-4|$ is positive and where it is negative, and the sketch in (ii) shows exactly that.

Solution

Part (i)

For $x<2$ we have $2-x>0$, so everything is defined. We differentiate the right-hand side, using the product rule on $-(2-x)\ln(2-x)$. The derivative of $-(2-x)$ is 1, and by the chain rule the derivative of $\ln(2-x)$ is $-\frac{1}{2-x}$. So

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\Big[-(2-x)\ln(2-x)+(2-x)\Big] &= 1\cdot\ln(2-x)-(2-x)\cdot\left(-\frac{1}{2-x}\right)-1\\ &= \ln(2-x)+1-1=\ln(2-x),\end{aligned}$$

and the stated antiderivative is correct.

(Alternatively, substitute $u=2-x$ and use $\int \ln u\,\mathrm{d}u = u\ln u-u$.)

In the same way, for $x>-2$,

$$\int \ln(2+x)\,\mathrm{d}x=(2+x)\ln(2+x)-(2+x)+c,$$

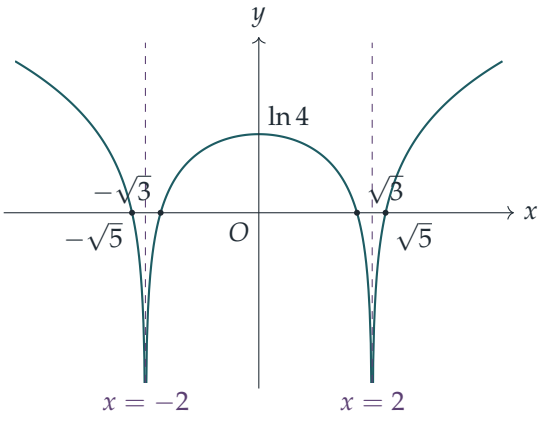
which we shall also need.

Part (ii)

Write $f(x)=\ln|x^2-4|$, defined for $x\neq\pm2$. Since $f(-x)=f(x)$, the curve is symmetric in the y-axis.

- For $|x| < 2$: $f(x) = \ln(4 - x^2)$, which is largest at $x = 0$, where $y = \ln 4$. It is zero when $4 - x^2 = 1$, i.e. $x = \pm\sqrt{3}$, and tends to $-\infty$ as $x \to \pm 2$.
- For $|x| > 2$: $f(x) = \ln(x^2 - 4)$, which rises from $-\infty$ near $x = \pm 2$, crosses the axis where $x^2 - 4 = 1$, i.e. $x = \pm\sqrt{5}$, and then grows slowly, like $2\ln|x|$.

The lines $x = \pm 2$ are vertical asymptotes.



Part (iii)

The finite region bounded by the positive x-axis, the y-axis and A is the arch between $x = 0$ and $x = \sqrt{3}$, where $y = \ln(4 - x^2) \geq 0$. Its area is

$$\int_0^{\sqrt{3}} \ln(4 - x^2)\,\mathrm{d}x = \int_0^{\sqrt{3}} \ln(2 - x)\,\mathrm{d}x + \int_0^{\sqrt{3}} \ln(2 + x)\,\mathrm{d}x,$$

using $\ln(4 - x^2) = \ln(2 - x) + \ln(2 + x)$, valid as both factors are positive here. By part (i),

$$\begin{aligned}\int_0^{\sqrt{3}} \ln(2 - x)\,\mathrm{d}x &= \Big[-(2 - x)\ln(2 - x) + (2 - x)\Big]_0^{\sqrt{3}} \\ &= -(2 - \sqrt{3})\ln(2 - \sqrt{3}) + (2 - \sqrt{3}) + 2\ln 2 - 2 \\ &= -(2 - \sqrt{3})\ln(2 - \sqrt{3}) - \sqrt{3} + 2\ln 2,\end{aligned}$$

and

$$\begin{aligned}\int_0^{\sqrt{3}} \ln(2 + x)\,\mathrm{d}x &= \Big[(2 + x)\ln(2 + x) - (2 + x)\Big]_0^{\sqrt{3}} \\ &= (2 + \sqrt{3})\ln(2 + \sqrt{3}) - (2 + \sqrt{3}) - 2\ln 2 + 2 \\ &= (2 + \sqrt{3})\ln(2 + \sqrt{3}) - \sqrt{3} - 2\ln 2.\end{aligned}$$

Ask yourself How do we combine $\ln(2 - \sqrt{3})$ with $\ln(2 + \sqrt{3})$?

Since $(2-\sqrt{3})(2+\sqrt{3}) = 4-3 = 1$, we have $\ln(2-\sqrt{3}) = -\ln(2+\sqrt{3})$. Adding the two integrals, the $\ln 2$ terms cancel and

$$\begin{aligned}\int_0^{\sqrt{3}} \ln(4-x^2)\,\mathrm{d}x &= (2-\sqrt{3})\ln(2+\sqrt{3}) + (2+\sqrt{3})\ln(2+\sqrt{3}) - 2\sqrt{3} \\ &= 4\ln(2+\sqrt{3}) - 2\sqrt{3},\end{aligned}$$

as required.

Part (iv)

For $|x| < 2$, B is the part of A between the asymptotes with the pieces below the axis reflected upwards. By the sketch, $\ln(4-x^2) \geq 0$ for $|x| \leq \sqrt{3}$ and $\ln(4-x^2) < 0$ for $\sqrt{3} < |x| < 2$. Using symmetry, the required area is

$$2\left[\int_0^{\sqrt{3}} \ln(4-x^2)\,\mathrm{d}x - \int_{\sqrt{3}}^{2} \ln(4-x^2)\,\mathrm{d}x\right].$$

The second integral is improper, because the integrand tends to $-\infty$ at $x = 2$. We integrate up to $x = 2-\varepsilon$ and let $\varepsilon \to 0^+$. In the antiderivative of $\ln(2-x)$ from (i), at $x = 2-\varepsilon$ we get

$$-\varepsilon\ln\varepsilon + \varepsilon \to 0,$$

by the given fact that $t\ln t \to 0$ as $t \to 0$. So the upper limit contributes 0 there, and

$$\begin{aligned}\int_{\sqrt{3}}^{2} \ln(2-x)\,\mathrm{d}x &= 0 - \left[-(2-\sqrt{3})\ln(2-\sqrt{3}) + (2-\sqrt{3})\right] \\ &= -(2-\sqrt{3})\ln(2+\sqrt{3}) - 2 + \sqrt{3},\end{aligned}$$

again using $\ln(2-\sqrt{3}) = -\ln(2+\sqrt{3})$. The other piece is an ordinary integral:

$$\begin{aligned}\int_{\sqrt{3}}^{2} \ln(2+x)\,\mathrm{d}x &= (4\ln 4 - 4) - \left[(2+\sqrt{3})\ln(2+\sqrt{3}) - (2+\sqrt{3})\right] \\ &= 8\ln 2 - 2 + \sqrt{3} - (2+\sqrt{3})\ln(2+\sqrt{3}).\end{aligned}$$

Adding,

$$\int_{\sqrt{3}}^{2} \ln(4-x^2)\,\mathrm{d}x = 8\ln 2 - 4 + 2\sqrt{3} - 4\ln(2+\sqrt{3}),$$

which is about -0.259: negative, as it should be. Therefore the area is

$$\begin{aligned}&2\Big[\big(4\ln(2+\sqrt{3}) - 2\sqrt{3}\big) - \big(8\ln 2 - 4 + 2\sqrt{3} - 4\ln(2+\sqrt{3})\big)\Big] \\ &\quad= 16\ln(2+\sqrt{3}) - 16\ln 2 + 8 - 8\sqrt{3} \\ &\quad= 16\ln\Big(\frac{2+\sqrt{3}}{2}\Big) + 8 - 8\sqrt{3} \approx 4.125.\end{aligned}$$

Checks

- Adding the two integrals over $[0, \sqrt{3}]$ and $[\sqrt{3}, 2]$ gives $\int_0^2 \ln(4 - x^2)\,\mathrm{d}x = 8\ln 2 - 4 \approx 1.545$. You can get this directly from (i) in one line, since at $x = 0$ the two antiderivatives give $-2\ln 2 + 2$ and $2\ln 2 - 2$, which cancel.
- The area in (iii) is $4\ln(2 + \sqrt{3}) - 2\sqrt{3} \approx 1.804$. That is less than the rectangle $\sqrt{3} \times \ln 4 \approx 2.40$ enclosing the arch, as it must be.
- Numerical integration of $\left|\ln|x^2 - 4|\right|$ over $(-2, 2)$ gives 4.1246, matching $16\ln\left(\frac{2+\sqrt{3}}{2}\right) + 8 - 8\sqrt{3} = 4.12457\ldots$.

Common slips

- Forgetting that the part of the curve between $x = \sqrt{3}$ and $x = 2$ lies below the axis, and so adding the signed integral instead of subtracting it.
- Leaving $\ln(2 - \sqrt{3})$ uncombined: the identity $(2 - \sqrt{3})(2 + \sqrt{3}) = 1$ is what makes the answers tidy.
- Simply substituting $x = 2$ into $(2 - x)\ln(2 - x)$ without comment: the expression is not defined there, and it is the limit $t \ln t \to 0$ that justifies the value 0.

Beyond STEP

The integral in (iv) is an *improper integral*: the integrand is unbounded, yet the area is finite because a logarithm blows up so slowly that $t \ln t \to 0$. At university you will make this precise, deciding exactly when $\int_0^1 g(t)\,\mathrm{d}t$ converges for unbounded g (for example, $\int_0^1 t^{-p}\,\mathrm{d}t$ converges exactly when $p < 1$). Abbott's *Understanding Analysis* treats limits like $t \ln t \to 0$ and the definition of the integral carefully.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.